

THE AI COMPUTE CAPITAL STACK

Capacity, signed pipeline, data-center atlas, cost curves and equity implications

Research-grade source-led assessment

Cut-off: 1 September 2026 | Market data: 31 August 2026 close

Core convention. Compute is deliverable IT accelerator power. Accelerator TDP, IT power and facility/utility power are kept separate. Capacity is normalized to H100-equivalent dense 8-bit throughput; memory, networking and software advantages remain outside the scalar. [S01-S11]

Anchor	Finding
Named public-site floor	13.3 GW IT / 14.6M H100e across 86 tracked campuses [S01]
Full-stack one-GW model	\$37.883B upfront / \$8.514B annual TCO [S03]
Largest modeled owner	Alphabet / Google: 5.04M H100e at Q4 2025 [S04,S05]
Largest explicit signed increments	Amazon/Anthropic up to 5 GW; Oracle/OpenAI 4.5 GW additional [S24,S33]

Prepared by OpenAI. Analytical research, not investment advice. Estimates are labeled; disclosures retain qualifiers such as “up to,” “more than,” and undisclosed overlap.

Executive summary

Central judgment. AI compute has become a capital-stack problem as much as a silicon problem. The decisive variables are power delivery, construction timing, lease/accounting structure, customer prepayments, contract tenor, utilization and refresh cadence. [S03,S12,S20,S23,S27,S31,S37]

- Google leads the comparable modeled fleet. The Q4 2025 chip-owner model estimates 5.04M H100e and 2.947 GW of accelerator TDP, implying approximately 4.72 GW IT under the central fleet bridge. Microsoft follows at 3.50M H100e, Amazon at 2.46M and Meta at 2.42M. [S04,S05; Model]
- The named-site database is a floor. It covers 86 campuses, 13.3 GW IT and 14.6M H100e, but estimates only 46% coverage of deployed global AI compute at end-June 2026. [S01,S02]
- A complete GB200-class GW is expensive: \$37.883B upfront and \$8.514B annual TCO. Facility shells are only one layer; powered-shell leases near \$1.4B-\$2.3B/GW-year are not comparable with full GPU services near \$9.7B-\$10.9B/GW-year. [S03,S17,S33,S34,S37,S39-S46; Model]
- Signed and contracted demand is larger than recognized assets. Amazon/Anthropic announced more than \$100B over ten years and up to 5 GW; Oracle/OpenAI announced 4.5 GW additional; Google described a 5 GW TPU program from 2027; CoreWeave disclosed 3.7 GW contracted. [S21,S22,S24,S33,S38]
- The footnotes are the real capacity map. Microsoft disclosed \$329.1B of mostly data-center uncommenced leases, Meta \$346.99B post-period, Oracle \$260B and CoreWeave \$35.5B. These figures are not directly additive with capex or purchase obligations. [S12,S27,S31,S37]
- Equity exposure bifurcates. Google has the best modeled compute-to-valuation combination among hyperscalers; Oracle and CoreWeave offer greater AI-capacity torque but carry much larger obligations relative to market capitalization. [S31,S37,S38,S48; Model]

Ticker	Current IT GW	Current H100e	Future indicator GW	Market cap	P/E	Capex / OCF	Uncommenced / MCap
AMZN	2.69	2.46M	5.00	\$2,832.3B	20.9x	137.8%	4.8%
MSFT	3.30	3.50M	0.66	\$3,776.8B	30.2x	63.4%	8.7%
GOOGL	4.72	5.04M	5.00	\$4,150.3B	17.0x	95.0%	2.1%
META	2.38	2.69M	5.00	\$1,468.6B	21.6x	79.5%	23.6%
ORCL	0.49	0.59M	4.50	\$434.2B	26.8x	174.1%	59.9%
CRWV	1.50	1.67M	2.20	\$46.8B	n/d	385.0%	75.9%
TSLA	0.20	0.23M	0.20	\$1,302.5B	340.7x	123.0%	n/d

Sources: current compute [S01,S04,S05,S37,S47; Model]; future [S16-S24,S28,S33,S38,S47]; financials [S12,S20,S23,S27,S31,S37,S47]; market data [S48].

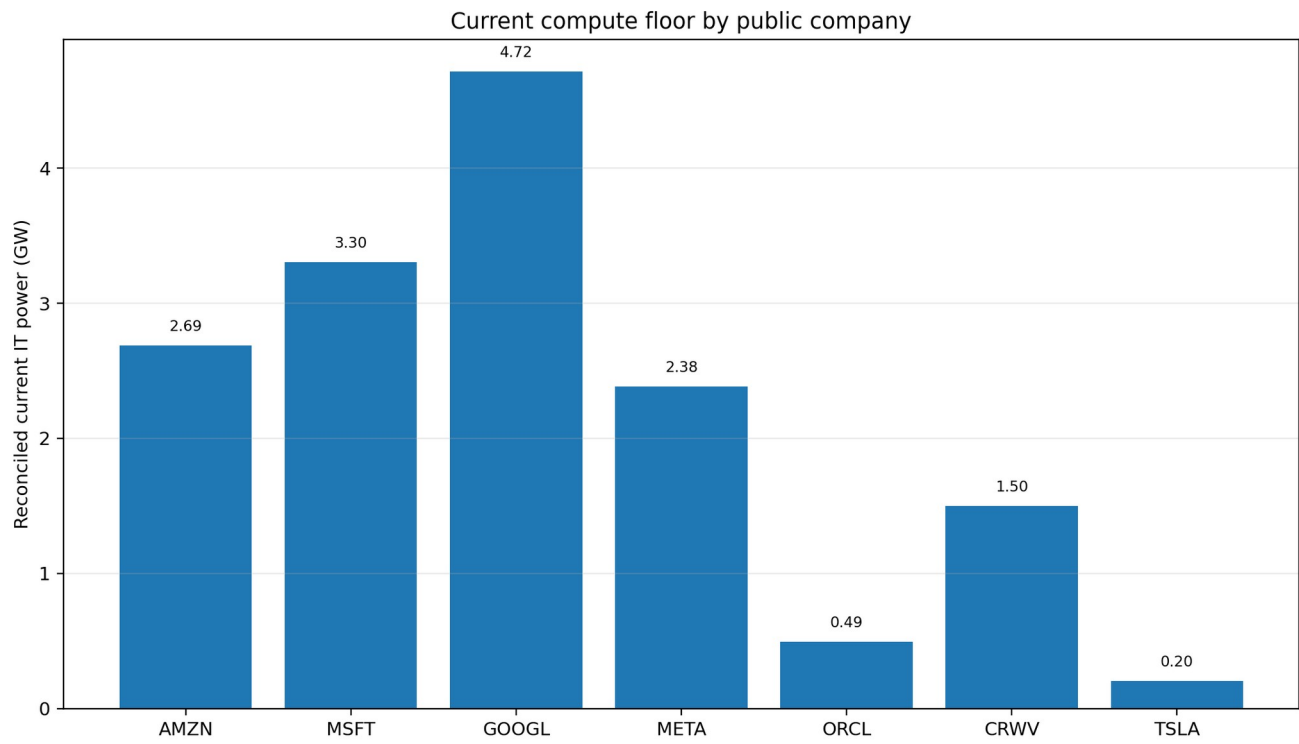


Figure 1. Reconciled current IT-power floor. Sources: [S01,S04,S05,S37,S47; Model].

Contents and source conventions

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Citation convention. Every material numerical claim carries a bracketed source ID. Derived values are marked “Model.” Site costs are modeled replacement costs, not company-reported project spend. Contract-value/GW ratios retain disclosed qualifiers and are not tariffs. [S01-S05]

Grade	Evidence	Treatment
A	SEC filing / audited annual report	Recognized assets, liabilities, cash capex and obligations; definitions still differ.
B	Company or named-counterparty announcement	High confidence for stated scope; overlap and timing may remain incomplete.
C	Independent dataset or model	Useful for comparability and site detail; uncertainty is material.
D	Derived normalization or ratio	Transparent arithmetic bounded by input assumptions.

1. Measurement framework

1.1 There is no single meaningful “gigawatt”

Accelerator TDP is the summed design power of chips. IT load includes host CPUs, memory, networking and storage. Facility load adds cooling, pumps, power conversion and redundancy. Utility or campus capacity can include spare blocks and non-AI loads. Treating these as interchangeable can overstate or understate comparable compute by 20%-100%. [S01,S02,S05-S11]

Convention	Includes	Best use	Main trap
Accelerator TDP	Chip design power only	Chip-owner model and silicon efficiency	Not deliverable facility capacity
IT power	Accelerators plus servers/fabric/storage	Preferred comparable data-center unit	Still depends on chip mix and utilization
Facility power	IT load multiplied by PUE	Utility draw and infrastructure burden	MW may be IT or gross
Campus / utility envelope	Interconnect or maximum planned load	Siting and grid context	May include redundancy, phases and general cloud

For fleet estimates based on accelerator TDP, the central bridge is facility power = 2.0x accelerator TDP and IT power = facility power / 1.25 PUE = 1.6x TDP. Sensitivity ranges are 1.7x-2.3x for facility/TDP and 1.18x-1.35x for PUE. [S05; Model]

1.2 H100-equivalent normalization

Platform	H100e / GPU	IT kW / GPU	Physical / IT GW	H100e / IT GW	H100e / facility GW	Source
H100 (DGX H100)	1.00	1.466	682,012	0.68M	0.57M	S05,S06
H200	1.00	1.466	682,012	0.68M	0.57M	S05,S06
B100 (modeled air-cooled envelope)	1.77	1.466	682,012	1.21M	1.01M	S05,S06,S09
B200 (DGX B200)	2.27	2.056	486,470	1.11M	0.92M	S05,S07,S08
GB200 NVL72	2.53	1.833	545,455	1.38M	1.17M	S05,S09,S10,S11

Interpretation. One IT GW can host roughly 682,000 H100s, 486,000 B200s or 545,000 GB200 GPUs under the modeled overhead assumptions. Comparable compute density rises from about 0.68M H100e/GW for H100 to 1.11M for B200 and 1.38M for GB200. [S05-S11; Model]

1.3 Counting and uncertainty rules

- Current named-site capacity is counted once at the attributed vendor roll-up, including dedicated third-party supply where the customer is named. [S01]
- Fleet chip-owner estimates are not added to named sites. Reconciled current compute takes the maximum of the site floor, chip-owner model and later direct disclosure. [S01,S04,S05; Model]
- Target-state sites are separate from signed contracts. A target may lack binding customer demand; a contract may be served across multiple sites. [S16-S34]
- The site database estimates 46% coverage and reports material uncertainty. This report uses floors and ranges rather than false precision. [S01,S02]

2. Current compute: comparable fleet view

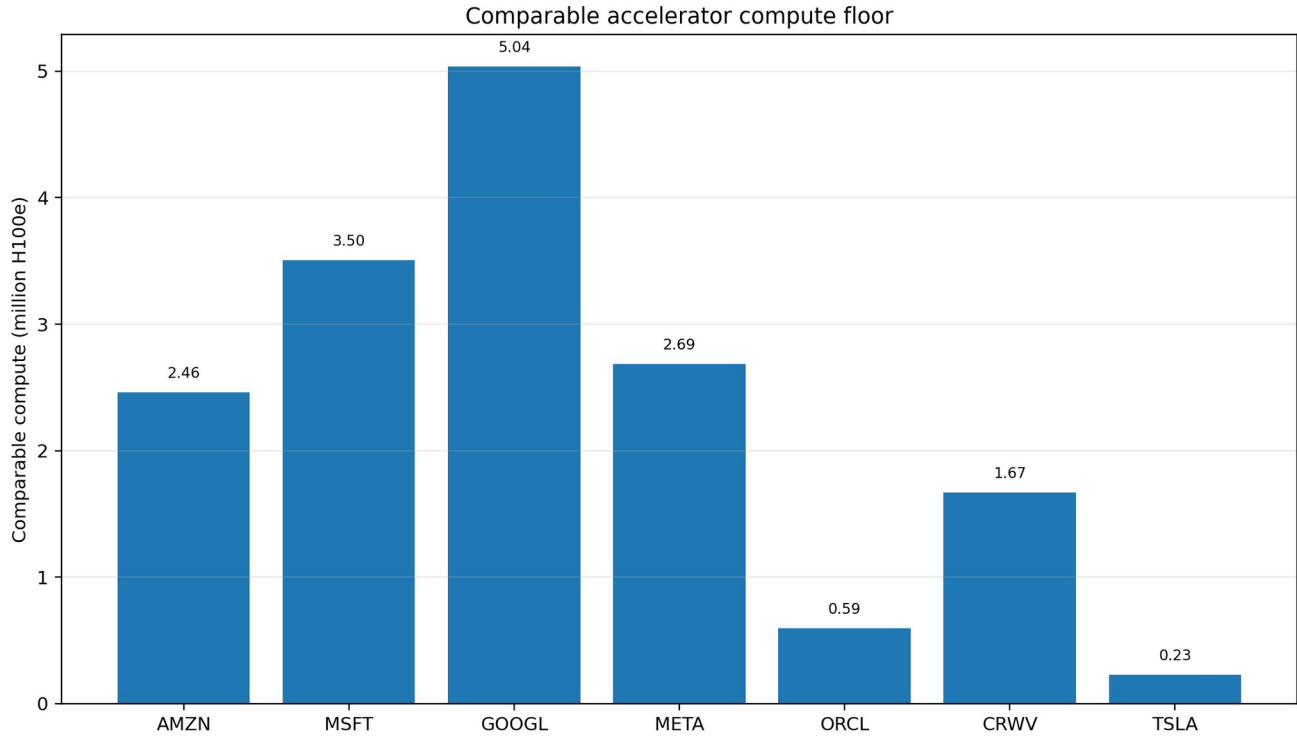


Figure 2. Reconciled current H100-equivalent compute. Sources: [S01,S04,S05,S37,S38,S47; Model].

2.1 Modeled full-fleet leadership

Owner	H100e	Accelerator TDP GW	Modeled IT GW	Modeled facility GW	Modeled mix
Alphabet / Google	5.04M	2.947	4.72	5.89	NVIDIA 24%, TPU 76%
Microsoft	3.50M	2.065	3.30	4.13	AMD 9%, NVIDIA 91%
Amazon / AWS	2.46M	1.679	2.69	3.36	Trainium 41%, NVIDIA 59%
Meta	2.42M	1.408	2.25	2.82	AMD 24%, NVIDIA 76%

Sources: [S04,S05]. IT/facility power use 1.6x and 2.0x accelerator TDP.

Google is the modeled leader at 5.04M H100e, driven by 3.81M TPU-equivalents plus 1.22M NVIDIA-equivalents. Microsoft follows at 3.50M, mostly NVIDIA. Amazon reaches 2.46M through 1.02M Trainium-equivalents and 1.44M NVIDIA-equivalents. Meta reaches 2.42M through NVIDIA and AMD. [S04,S05]

2.2 Reconciled public-company floor

Ticker	Named sites IT GW	Named sites H100e	Reconciled IT GW	Reconciled H100e	Source
AMZN	1.74	1.35M	2.69	2.46M	S01,S04,S05,S23,S24
MSFT	1.57	1.74M	3.30	3.50M	S01,S04,S05,S12,S16-S19
GOOGL	3.30	4.61M	4.72	5.04M	S01,S04,S05,S20-S22
META	2.38	2.69M	2.38	2.69M	S01,S04,S05,S27-S29
ORCL	0.49	0.59M	0.49	0.59M	S01,S31-S34
CRWV	0.68	0.76M	1.50	1.67M	S01,S37,S38; Model
TSLA	0.00	0.00M	0.20	0.23M	S47; Model

For Google, Microsoft and Amazon, the Q4 2025 fleet model exceeds the named-site floor, consistent with legacy and smaller-region capacity outside the database. Meta is the reverse: its later named-site floor exceeds the older chip-owner model. CoreWeave uses its disclosed 1.5 GW active power, while Tesla uses a direct cluster floor above 205 MW.
[S01,S04,S05,S37,S47; Model]

3. Data-center atlas: current and target sites

3.1 Current named-site roll-up

Vendor roll-up	IT GW	H100e	Modeled replacement cost	H100e k / IT MW	Source
Alphabet / Google	3.30	4.61M	\$125.0B	1.40	S01,S02
Meta	2.38	2.69M	\$90.2B	1.13	S01,S02
Amazon / AWS	1.74	1.35M	\$66.0B	0.78	S01,S02
Microsoft	1.57	1.74M	\$59.4B	1.11	S01,S02
xAI / SpaceXAI	0.96	1.12M	\$36.5B	1.17	S01,S02
CoreWeave	0.68	0.76M	\$25.7B	1.11	S01,S02
Oracle	0.49	0.59M	\$18.7B	1.20	S01,S02
Anthropic	0.04	0.06M	\$1.6B	1.40	S01,S02

Google has the largest identifiable site floor at approximately 3.30 GW IT and 4.61M H100e. Meta has 2.38 GW and 2.69M. Amazon, including dedicated Anthropic capacity, has 1.74 GW and 1.35M. Microsoft has 1.57 GW and 1.74M including a 50 MW Nebius site attributed to dedicated supply. [S01,S02; Model roll-up]

3.2 Twenty largest current sites

Owner / customer	Site	IT MW	H100e	Modeled cost	Source
xAI / SpaceXAI	Colossus 2	946	1,112k	\$35.84B	S01,S02
Amazon / Anthropic	New Carlisle	910	686k	\$34.47B	S01,S02
Microsoft	Fairwater Atlanta	636	769k	\$24.09B	S01,S02
Meta	Prometheus	562	680k	\$21.29B	S01,S02
Google	New Albany	453	616k	\$17.16B	S01,S02
Oracle / OpenAI	Stargate Abilene	421	509k	\$15.95B	S01,S02
Microsoft	Fairwater Wisconsin	369	446k	\$13.98B	S01,S02
Google	Pryor North	368	763k	\$13.94B	S01,S02
Google	Columbus	303	409k	\$11.48B	S01,S02
Amazon	Madison	284	214k	\$10.76B	S01,S02
CoreWeave	Denton	262	253k	\$9.93B	S01,S02
Google	Bristow	238	239k	\$9.02B	S01,S02
Google	Council Bluffs East	237	289k	\$8.98B	S01,S02
Google	Omaha	237	268k	\$8.98B	S01,S02
Google	Papillion	237	80k	\$8.98B	S01,S02
Amazon	Ridgeland	228	171k	\$8.64B	S01,S02
Amazon	New Albany	213	201k	\$8.07B	S01,S02
Microsoft	Goodyear	202	205k	\$7.65B	S01,S02
Microsoft	Project Osmium	190	156k	\$7.20B	S01,S02
Google	Mesa	183	343k	\$6.93B	S01,S02

Concentration. The five largest tracked live sites total approximately 3.51 GW IT, about 26% of the database. They span xAI, Amazon/Anthropic, Microsoft, Meta and Google. [S01; Model]

3.3 Planned and target-state roll-up

Vendor / sponsor	Target-state IT GW	Incremental IT GW	Incremental H100e	Modeled target cost	Source
OpenAI	9.99	9.99	20.25M	\$269.4B	S01,S02
Meta	4.33	3.31	4.96M	\$124.5B	S01,S02
Amazon / AWS	4.75	3.01	2.15M	\$149.8B	S01,S02
Microsoft	2.84	2.47	4.32M	\$89.2B	S01,S02
Alphabet / Google	3.20	2.28	6.83M	\$100.1B	S01,S02
QTS	1.59	1.59	3.70M	\$42.9B	S01,S02
CoreWeave	2.09	1.55	1.15M	\$55.9B	S01,S02
xAI / SpaceXAI	2.06	1.12	0.69M	\$58.0B	S01,S02
Crusoe	0.94	0.94	1.70M	\$25.5B	S01,S02
Oracle	1.18	0.76	0.49M	\$31.9B	S01,S02
Anthropic	0.74	0.70	0.76M	\$22.2B	S01,S02

The selected planned-site inventory is dominated by OpenAI/Stargate at approximately 9.99 GW of incremental target capacity, followed by Meta at 3.31 GW, Amazon at 3.01 GW, Microsoft at 2.47 GW and Google at 2.28 GW. These are target estimates and not necessarily signed. [S01,S02; Model matching]

3.4 Largest target-state campuses

Owner / sponsor	Site	Target IT MW	Incremental MW	Target H100e	Modeled cost	Source
Microsoft	Fairwater Wisconsin	2,715	2,346	4,500k	\$85.73B	S01,S02
OpenAI	Stargate New Mexico	2,450	2,450	5,100k	\$66.29B	S01,S02
Amazon / Anthropic	New Carlisle	2,310	1,400	1,700k	\$72.92B	S01,S02
Meta	Hyperion	2,262	2,262	4,300k	\$63.49B	S01,S02
xAI / SpaceXAI	Colossus 2	2,065	1,119	1,800k	\$58.00B	S01,S02
OpenAI	Stargate Shackelford	1,960	1,960	4,600k	\$53.03B	S01,S02
QTS	Cedar Rapids	1,587	1,587	3,700k	\$42.92B	S01,S02
OpenAI	Stargate UAE	1,400	1,400	2,000k	\$37.88B	S01,S02
OpenAI	Stargate Michigan	1,383	1,383	2,900k	\$37.43B	S01,S02
OpenAI	Stargate Wisconsin	1,300	1,300	2,600k	\$34.17B	S01,S02
OpenAI	Stargate Milam	1,200	1,200	2,500k	\$32.46B	S01,S02
Oracle / OpenAI	Stargate Abilene	1,180	759	1,000k	\$31.93B	S01,S02
Meta	Prometheus	1,110	548	1,000k	\$32.35B	S01,S02
Google	Fort Wayne	1,035	1,035	3,600k	\$32.69B	S01,S02
Amazon	Ridgeland	1,008	780	788k	\$31.82B	S01,S02
Crusoe	Abilene expansion	941	941	1,700k	\$25.46B	S01,S02
Amazon	Madison	819	535	514k	\$25.87B	S01,S02
CoreWeave	Helios	800	668	638k	\$20.00B	S01,S02
CoreWeave	Ellendale	520	452	505k	\$15.15B	S01,S02
Google	Omaha	474	237	1,100k	\$14.96B	S01,S02

Targets above 2 GW IT include Microsoft Fairwater Wisconsin, OpenAI Stargate New Mexico, Amazon/Anthropic New Carlisle, Meta Hyperion and xAI Colossus 2. Epoch assigns 80% confidence intervals of roughly 1.4x for site IT power, 1.5x for compute and 1.6x for cost. [S01,S02]

4. Signed and contracted pipeline

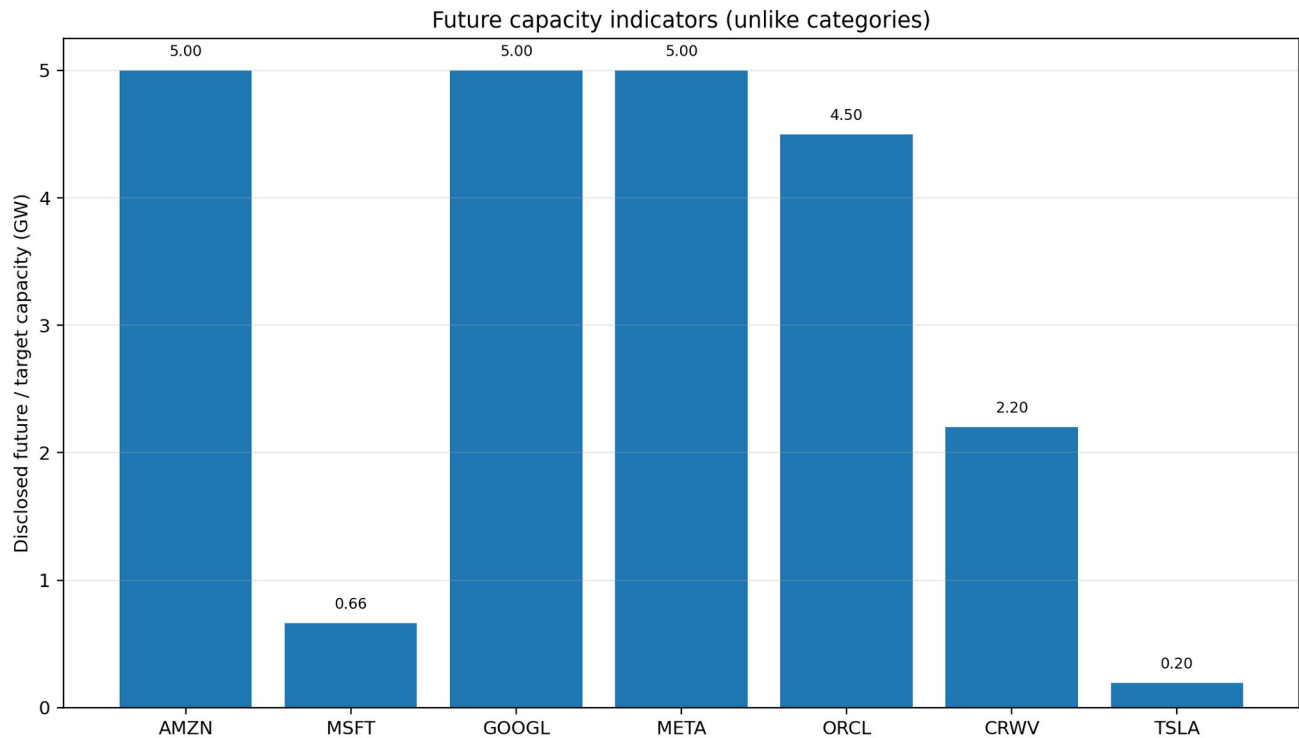


Figure 3. Future capacity indicators. Categories are not additive. Sources: [S16-S24,S28,S33,S38,S47].

Vendor	Counterparty	GW	Category	Value	Value / GW-year	Counting note	Source
Amazon / AWS	Anthropic	5.00	Signed / announced	\$100.0B	\$2.0B	Upper bound; named site targets are not additive without mapping.	S24
Alphabet / Google	Anthropic	1.00	Signed / announced	n/d	n/d	May be part of 5 GW program; do not sum.	S21
Alphabet / Google	Anthropic / Broadcom program	5.00	Signed / announced	n/d	n/d	Overlap with 2026 milestone undisclosed.	S22
Microsoft	Anthropic	1.00	Demand commitment	\$30.0B	n/d	May be served by outsourced suppliers below; do not add.	S19
Microsoft	IREN	0.20	Firm supply contract	\$9.7B	\$9.7B	Specific outsourced capacity; \$5.8B hardware capex separately.	S17
Microsoft	Nscale	0.46	Signed supply; partly modeled	n/d	n/d	Applies Texas MW/GPU ratio; 0.7 GW option excluded.	S16
Microsoft	Nebius	n/d	Firm supply contract	\$17.4B	n/d	No defensible GW derivation.	S18
Oracle	OpenAI	4.50	Signed / announced	\$300.0B	\$13.3B	Use 4.5 GW incremental; 5.5 GW only for a rough cross-announcement ratio.	S33,S34
CoreWeave	Multiple customers	3.70	Company disclosed	\$104.0B	n/d	Backlog duration/service mix vary.	S38
Meta	Internal demand	5.00	Project target	\$50.0B	n/d	Not a customer contract; capacity/investment scopes may differ.	S28,S29

Vendor	Counterparty	GW	Category	Value	Value / GW-year	Counting note	Source
Tesla	Internal AI	0.40	Company roadmap estimate	n/d	n/d	Visual estimate; current direct cluster floor >205 MW.	S47

4.1 Amazon / Anthropic

Anthropic and Amazon announced an expansion above \$100B over ten years for up to 5 GW of new AWS capacity. Nearly 1 GW of Trainium2/Trainium3 capacity is expected by end-2026, and Anthropic stated that more than one million Trainium2 chips were already in use. The hybrid ratio is above \$2B/GW-year, but it is not a tariff because value is a floor, capacity an upper bound and the ramp undisclosed. [S24; Model]

Project Rainier was fully operational in June 2025 with nearly 500,000 Trainium2 chips and an expected expansion above one million. New Carlisle is the second-largest tracked live AI site at 910 MW IT. [S01,S25]

4.2 Google / Anthropic

Anthropic announced up to one million Google TPUs, tens of billions of dollars of services and well over 1 GW online in 2026. A separate announcement describes a 5 GW Google/Broadcom program beginning in 2027. Because overlap is undisclosed, this report shows 1 GW as a milestone and 5 GW as the program envelope rather than summing them. [S21,S22]

4.3 Microsoft: demand versus supply

Microsoft and Anthropic announced a \$30B Azure purchase commitment and up to 1 GW of additional capacity. Separately, Microsoft contracted 200 MW IT from IREN for \$9.7B over five years, approximately 200,000 GB300 GPUs from Nscale, and a \$17.4B five-year GPU-services agreement with Nebius whose MW is undisclosed. Applying Nscale Texas disclosure of 240 MW for 104,000 GPUs to the total GPU count implies approximately 462 MW IT; a 700 MW option is excluded. [S16-S19; Model]

Do not double count. The 1 GW Anthropic demand commitment may be served by some combination of owned Fairwater assets and outsourced IREN, Nscale or Nebius capacity. Supply and demand rows are therefore separate. [S13-S19]

4.4 Oracle / OpenAI and CoreWeave

OpenAI announced 4.5 GW of additional Oracle capacity, bringing the Oracle-linked program above 5 GW under development and above two million chips. A linked announcement referenced more than \$300B and more than 5.5 GW; dividing those values produces a rough \$10.9B/GW-year over five years, but it is not a clean tariff. [S33,S34; Model]

CoreWeave disclosed 1.5 GW active and 3.7 GW contracted, so the simple incremental bridge is 2.2 GW. Its \$104B backlog divided by contracted capacity is \$28.1B/GW, but duration and service mix vary. [S38; Model]

5. Footnote archaeology and capital commitments

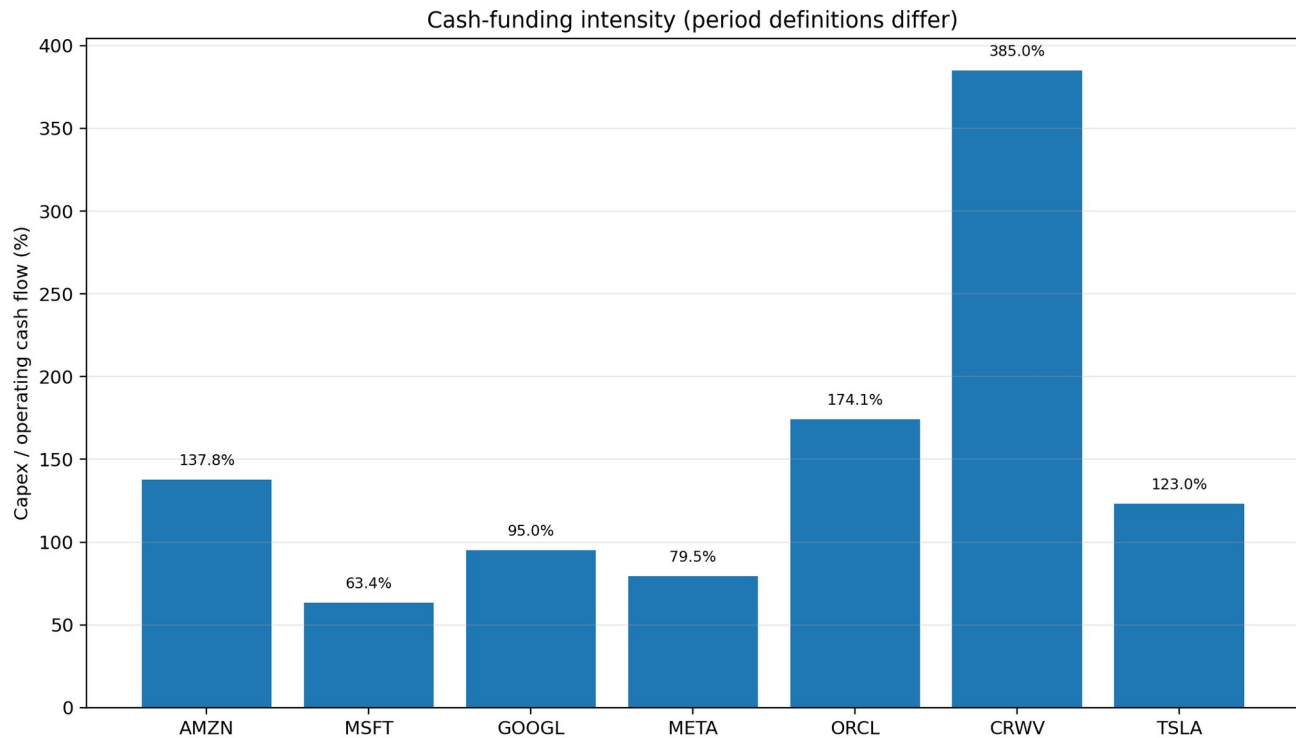


Figure 4. Capex relative to operating cash flow. Period definitions differ. Sources: [S12,S20,S23,S27,S31,S37,S47].

Ticker	Period	Capex	OCF	Capex / OCF	Net PPE	CIP / not in service	Uncommenced leases	RPO / backlog	Source
AMZN	H1 2026 / Q2 BS	\$98.41B	\$71.42B	137.8%	\$263.75B	n/d	\$137.21B	\$496.00B	S23
MSFT	FY2026	\$115.95B	\$182.94B	63.4%	\$313.08B	n/d	\$329.10B	\$684.00B	S12
GOOGL	H1 2026 / Q2 BS	\$80.60B	\$84.86B	95.0%	\$321.21B	\$122.81B	\$85.20B	\$519.50B	S20
META	H1 2026 / Q2 BS	\$50.92B	\$64.09B	79.5%	\$225.72B	\$80.34B	\$346.99B	n/d	S27
ORCL	FY2026	\$55.66B	\$31.98B	174.1%	\$99.96B	\$39.97B	\$260.00B	\$638.00B	S31,S32
CRWV	H1 2026 / Q2 BS	\$14.12B	\$3.66B	385.4%	\$46.74B	\$11.92B	\$35.50B	\$104.00B	S37,S38
TSLA	Q2 2026	\$5.79B	\$4.70B	123.2%	n/d	n/d	n/d	n/d	S47

Cash capex understates asset formation when finance leases, OEM financing, prepayments or JVs fund the build. Gross obligations can overstate AI exposure because they may include fulfillment, office, content, energy and general cloud assets. [S12,S20,S23,S27,S31,S37]

5.1 Microsoft

Microsoft reported FY2026 additions to PPE of \$115.948B, net PPE of \$313.076B and construction commitments of \$34.566B. Total disclosed obligations were \$743.821B, including \$443.506B of leases including interest and \$194.060B of purchase commitments. Uncommenced leases, mostly data centers, were \$329.1B and begin FY2027-FY2033 with 1-20 year terms. RPO was \$684B. [S12]

5.2 Alphabet

Alphabet spent \$80.598B of first-half capex against \$84.859B of OCF. Net PPE was \$321.212B, including \$122.814B of assets not yet in service. Uncommenced leases were \$85.2B, primarily data centers starting in 2026-2031. Purchase commitments and other obligations were \$811.0B; the filing also disclosed \$24.1B of estimated data-center/energy backstops and \$20B of private-company capital commitments. Cloud RPO was \$513.9B. [S20]

5.3 Amazon

Amazon reported \$98.411B of first-half cash PPE purchases against \$71.419B of OCF. AWS net PPE was \$263.750B and first-half AWS net additions were \$90.120B. AWS RPO was \$496B with a 6.4-year average duration. Amazon disclosed \$137.214B of uncommenced leases and \$650.034B of total commitments, but these totals include non-data-center categories. [S23]

5.4 Meta

Meta reported first-half capex including finance-lease principal of \$50.920B against \$64.088B of OCF. Net PPE was \$225.724B and construction in progress \$80.345B. Uncommenced leases were \$278.99B at quarter-end; another \$68B of data-center leases signed in July took the post-period figure to \$346.99B. Contractual commitments were \$349.31B; the totals likely overlap and are not added. [S27]

The Blue Owl Hyperion JV is an 80/20 structure covering approximately \$27B of long-lived facility assets. Blue Owl contributed approximately \$7B cash, Meta received a \$3B distribution and Meta leases the facilities with a 16-year residual-value guarantee. [S29]

5.5 Oracle

Oracle FY2026 capex was \$55.663B against \$31.977B of OCF. Customer prepayments with a financing component contributed \$4.592B to OCF; short-term capex financing added \$3.345B and debt issuance was \$46.093B. Net PPE was \$99.957B and construction in progress \$39.973B. [S31]

Oracle disclosed \$260B of uncommenced leases, substantially all data centers, beginning FY2027-FY2029 with 15-19 year terms and including a \$3.3B lessor guarantee. Other purchase obligations were \$13.309B. RPO reached \$638B. [S31,S32]

5.6 CoreWeave

CoreWeave reported net PPE of \$46.736B and operating-lease ROU assets of \$16.595B. First-half cash PPE purchases were \$14.117B, while \$9.796B of PPE liabilities/OEM financing and \$8.359B of ROU additions were noncash. Uncommenced leases were \$35.5B; one undelivered 393 MW site carried up to \$14.7B of payments over 16 years. Debt principal was \$35.551B. [S37]

CoreWeave had \$103.7B of unsatisfied RPO in the 10-Q and approximately \$104B of reported backlog; 1.5 GW active and 3.7 GW contracted. Its three largest Q2 customers represented 36%, 26% and 10% of revenue. [S37,S38]

6. Vendor dossiers

6.1 Amazon / AWS

Metric	Value	Evidence / interpretation
Reconciled current	2.69 GW / 2.46M	Q4 2025 model exceeds named-site floor [S01,S04,S05; Model]
Named-site floor	1.74 GW	Includes New Carlisle 910 MW [S01]
Signed pipeline	Up to 5 GW / >\$100B over 10 years	Nearly 1 GW by end-2026 [S24]
Equity	\$259.77 / \$2,832.3B / 20.9x	31 Aug. 2026 [S48]

AWS has the strongest vertically integrated named anchor: customer, ten-year minimum value, up-to-5-GW envelope and in-house Trainium roadmap. Trainium is approximately 41% of modeled Amazon H100e. [S04,S05,S24,S25]

The cost is cash intensity. First-half cash PPE exceeded OCF by 38%, and AWS added \$90.120B of net PPE in six months. The \$15B Indiana / 2.4 GW headline is \$6.25B/GW at campus/infrastructure scope, not full stack. [S23,S26; Model]

Stock implication. At 20.9x, Amazon is not the most expensive hyperscaler despite the 5 GW anchor. The key test is whether Trainium lowers delivered cost enough to offset near-term negative free cash flow and whether the Anthropic ramp fills assets before refresh. [S23-S25,S48]

6.2 Microsoft

Metric	Value	Evidence / interpretation
Reconciled current	3.30 GW / 3.50M	Q4 2025 owner model [S04,S05; Model]
Named-site floor	1.57 GW	Fairwater and other sites [S01,S13-S15]
Firm supplier capacity	Approximately 0.662 GW plus Nebius	IREN 200 MW; Nscale modeled 462 MW [S16-undisclosed S18]
Equity	\$507.29 / \$3,776.8B / 30.2x	31 Aug. 2026 [S48]

Microsoft combines a large owned fleet with a broad outsourced network. IREN, Nscale and Nebius shift some construction, power and hardware financing to suppliers while reserving capacity for Azure. [S13-S18]

The benefit is speed and diversification; the cost is margin sharing and a complex stack. Microsoft had \$329.1B of mostly data-center uncommenced leases and \$743.821B of obligations, alongside \$684B of RPO. [S12]

Stock implication. Microsoft has the best broad cloud demand visibility but trades at 30.2x, the highest P/E among the four hyperscalers. Its premium requires disciplined economics across owned and outsourced capacity. [S12,S16-S19,S48]

6.3 Alphabet / Google

Metric	Value	Evidence / interpretation
Reconciled current	4.72 GW / 5.04M	Largest modeled fleet [S04,S05; Model]
Named-site floor	3.30 GW	Largest site roll-up [S01]
Pipeline	5 GW program from 2027; >1 GW 2026 milestone	Overlap undisclosed [S21,S22]
Equity	\$339.35 / \$4,150.3B / 17.0x	31 Aug. 2026 [S48]

Google is the compute leader under both the chip-owner model and the site floor. Approximately 76% of modeled H100e comes from TPUs, creating the strongest custom-silicon position. H100e does not capture TPU workload fit or fungibility. [S01,S04,S05]

Alphabet has \$122.814B of assets not in service, versus \$247.177B of technical infrastructure in service. Cloud RPO is \$513.9B, while the \$811B broad commitment total makes the forward envelope less transparent. [S20]

Stock implication. Google combines the largest modeled compute base with the lowest P/E of the four hyperscalers at 17.0x. This is the strongest headline compute-to-valuation setup, subject to sustained TPU utilization. [S04,S05,S20,S48]

6.4 Meta

Metric	Value	Evidence / interpretation
Reconciled current	2.38 GW / 2.69M	Named-site floor exceeds older model [S01,S04,S05]
Named-site floor	2.38 GW	Prometheus 562 MW is largest [S01]
Target	Hyperion 5 GW / >\$50B regional investment	Internal target [S28]
Equity	\$572.34 / \$1,468.6B / 21.6x	31 Aug. 2026 [S48]

Meta monetizes compute through recommendation, advertising and model capability rather than external cloud contracts. There is no independently priced tariff to validate unit economics. [S27,S28]

The balance-sheet strategy mixes direct construction, leases and Blue Owl JV financing. The raw \$27B / 5 GW = \$5.4B/GW ratio covers long-lived facilities, not servers, and scope/timing are mismatched. [S28,S29; Model]

Stock implication. Meta trades at 21.6x with strong ads cash generation, but post-period uncommenced leases equal 23.6% of market cap. Investors underwrite internal monetization plus long-lived lease and residual commitments. [S27-S29,S48; Model]

6.5 Oracle

Metric	Value	Evidence / interpretation
Current named-site floor	0.49 GW / 0.59M	Abilene and Batam [S01]
Signed pipeline	4.5 GW additional Oracle/OpenAI	>5 GW including Abilene under development [S33]
RPO / market cap	146.9%	\$638.0B RPO / \$434.2B market cap [S32,S48; Model]
Equity	\$149.12 / \$434.2B / 26.8x	31 Aug. 2026 [S48]

Oracle has the most concentrated public AI demand bet. The OpenAI program adds 4.5 GW and a linked >\$300B commitment, creating exceptional visibility and exceptional sensitivity to execution and financing. [S31-S34]

FY2026 capex was 174% of OCF. Customer prepayments and debt funded the build, while \$260B of uncommenced data-center leases equals approximately 60% of market cap. [S31,S48; Model]

Stock implication. Oracle offers the highest upside torque if capacity comes online on time and RPO converts at attractive margins, and the sharpest downside if power, construction, customer timing or financing moves adversely. [S31-S34,S48]

6.6 CoreWeave

Metric	Value	Evidence / interpretation
Active / contracted	1.5 GW / 3.7 GW	2.2 GW simple incremental [S38; Model]
Modeled current	1.67M H100e	Active power times observed density [S01,S38; Model]
Backlog / market cap	222.3%	\$104.0B / \$46.8B [S38,S48; Model]
Equity	\$84.89 / \$46.8B / P/E n/m	31 Aug. 2026 [S48]

CoreWeave is the purest public compute exposure; its landlord network reveals powered-shell pricing that hyperscaler filings obscure. [S37-S46]

Debt and uncommenced leases each equal roughly three-quarters of market cap; net PPE approximates market cap; and three customers represented 72% of Q2 revenue. Short hardware life amplifies utilization and renewal risk. [S37,S38,S48; Model]

Stock implication. Backlog can drive rapid revenue growth, but market cap per current IT GW of \$31.2B is meaningful only after recognizing debt, leases, refresh capex and customer concentration. [S37,S38,S48; Model]

6.7 Tesla

Metric	Value	Evidence / interpretation
Current cluster floor	>205 MW	Cortex 1 >90 MW; Cortex 2 >115 MW [S47]
Comparable compute	>230k H100e	Company roadmap-equivalent [S47]
Near-term target	Approximately 400 MW by Dec. 2026	Visual estimate, not a cloud contract [S47; Model]
Equity	\$367.95 / \$1,302.5B / 340.7x	31 Aug. 2026 [S48]

Tesla compute supports autonomy, robotics and model development; it is not sold as general-purpose cloud. Q2 company-wide capex of \$5.789B exceeded OCF of \$4.697B, but compute is only one capital use. [S47]

At \$37.883B/GW, 205 MW implies approximately \$7.8B of replacement capex, below 1% of market cap. A 400 MW target is approximately \$15.2B, around 1.2% of market cap. [S03,S47,S48; Model]

Stock implication. Tesla's market cap per current IT GW exceeds \$6.3T, demonstrating the limitation of asset comparisons. Compute is an enabling strategic option rather than primary asset backing. [S47,S48; Model]

7. Counterparty chain and market pricing

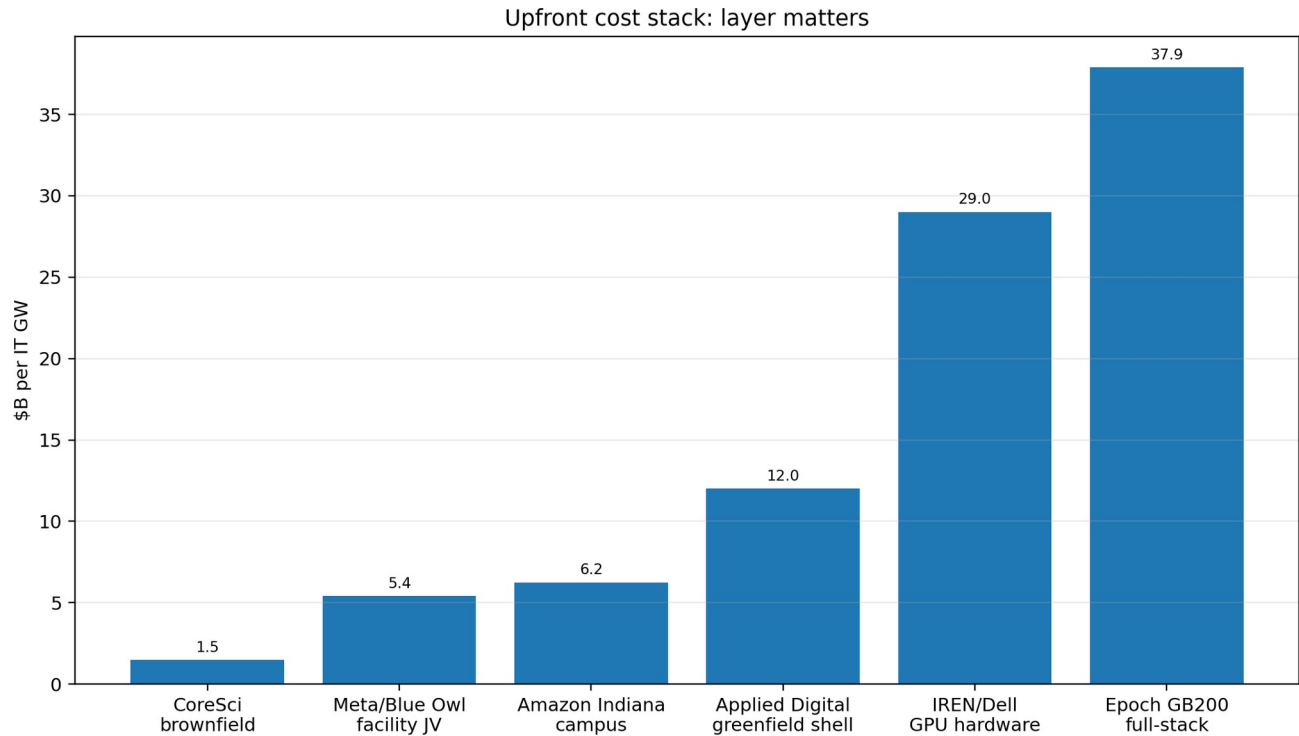


Figure 5. Upfront cost layers per comparable IT GW. Sources: [S03,S17,S26,S28,S29,S40,S46; Model].

Hyperscaler filings aggregate data centers with broader PPE and commitments. Counterparty disclosures provide the missing denominator: critical IT MW, term, contract value, power pass-through and buildout responsibility. [S16-S19,S29,S37,S39-S46]

7.1 Powered-shell lease market

Benchmark	MW	Years	Value	Annual \$B/GW	\$/kW-month	Power	Source
Core Scientific / CoreWeave	590	12	\$10.20B	\$1.44B	\$120	No	S39,S40
Applied Digital / CoreWeave 400 MW	400	15	\$11.00B	\$1.83B	\$153	Pass-through	S43,S46
Applied Digital 200 MW AI factory	200	15	\$5.00B	\$1.67B	\$139	Pass-through	S44
Galaxy Helios / CoreWeave	526	15	\$15.00B	\$1.90B	\$158	No / pass-through	S42
CoreWeave significant undelivered site	393	16	\$14.70B	\$2.34B	\$195	Unclear	S37

The disclosed powered-shell range is approximately \$120-\$195/kW-month, or \$1.44B-\$2.34B/GW-year, generally excluding tenant GPU hardware and often passing through electricity. [S37,S39-S44; Model]

7.2 Full service and customer commitments

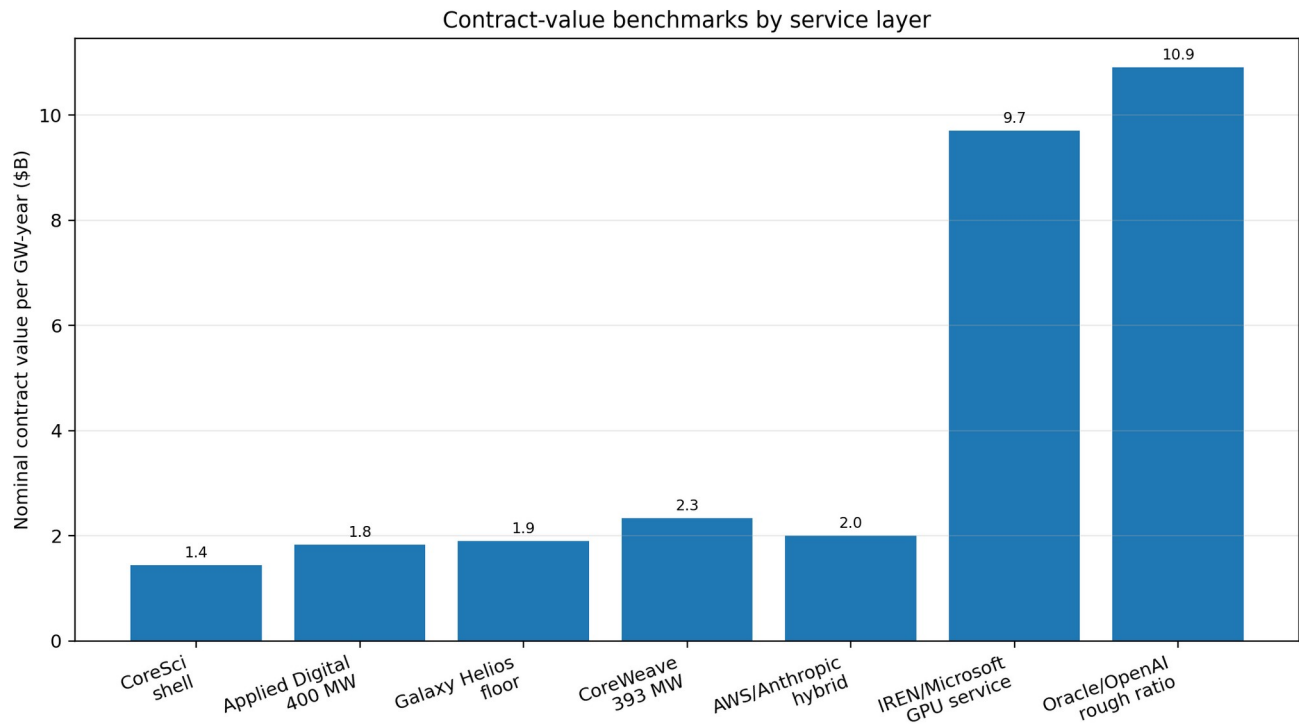


Figure 6. Contract-value benchmarks per GW-year. Sources: [S17,S24,S33,S34,S37,S39-S44; Model].

IREN's 200 MW Microsoft contract is approximately \$808/kW-month and \$9.7B/GW-year for a full GPU service, versus \$29B/GW of separately disclosed Dell hardware capex. The spread over powered-shell pricing reflects accelerators, refresh, operations, financing and service margin. [S17; Model]

A rough Oracle/OpenAI cross-announcement ratio is \$10.9B/GW-year, slightly above the \$8.514B/GW annual TCO reference. It is illustrative because the numerator and denominator come from linked rather than identical disclosures. [S03,S33,S34; Model]

7.3 Financing migration

Buyer / tenant	Counterparties	Structure	Benefit	Residual risk
Microsoft	IREN / Nscale / Nebius	Supplier funds sites and/or GPUs; Microsoft buys capacity	Speed and optionality	Supplier margin and long commitments [S16-S18]
Meta	Blue Owl JV	JV owns long-lived facility assets; Meta leases all capacity	Shared construction capital	Lease and residual guarantee remain [S29]
CoreWeave	Core Scientific / Galaxy / Applied Digital	Landlords build shells; CoreWeave finances IT	Critical-MW denominators disclosed	Debt/lease and utilization risk at tenant [S37,S39-S46]
Oracle	OpenAI prepayments / debt / lessors	Customer financing and debt support build	RPO and funding visibility	Counterparty concentration and long leases [S31-S34]

8. One-GW cost curves

8.1 Published base case

Component	Upfront	Annualized / opex
Servers	\$21.188B	\$5.021B
Facility	\$11.433B	\$1.387B
Network	\$4.925B	\$1.167B
Land	\$0.172B	\$0.013B
Utility interconnect	\$0.164B	\$0.020B
Energy	-	\$0.594B
Taxes / maintenance / labor / water / other	-	\$0.312B
Total	\$37.883B	\$8.514B

Epoch AI one-GW GB200-class model [S03]. Five-year IT life and 14-year facility life.

Servers and network are 69% of upfront cost and about 73% of annualized TCO. Extending useful IT life from three to seven years lowers annual TCO from approximately \$12B to \$7B/GW; hardware refresh dominates land and power. [S03]

8.2 Utilization and refresh

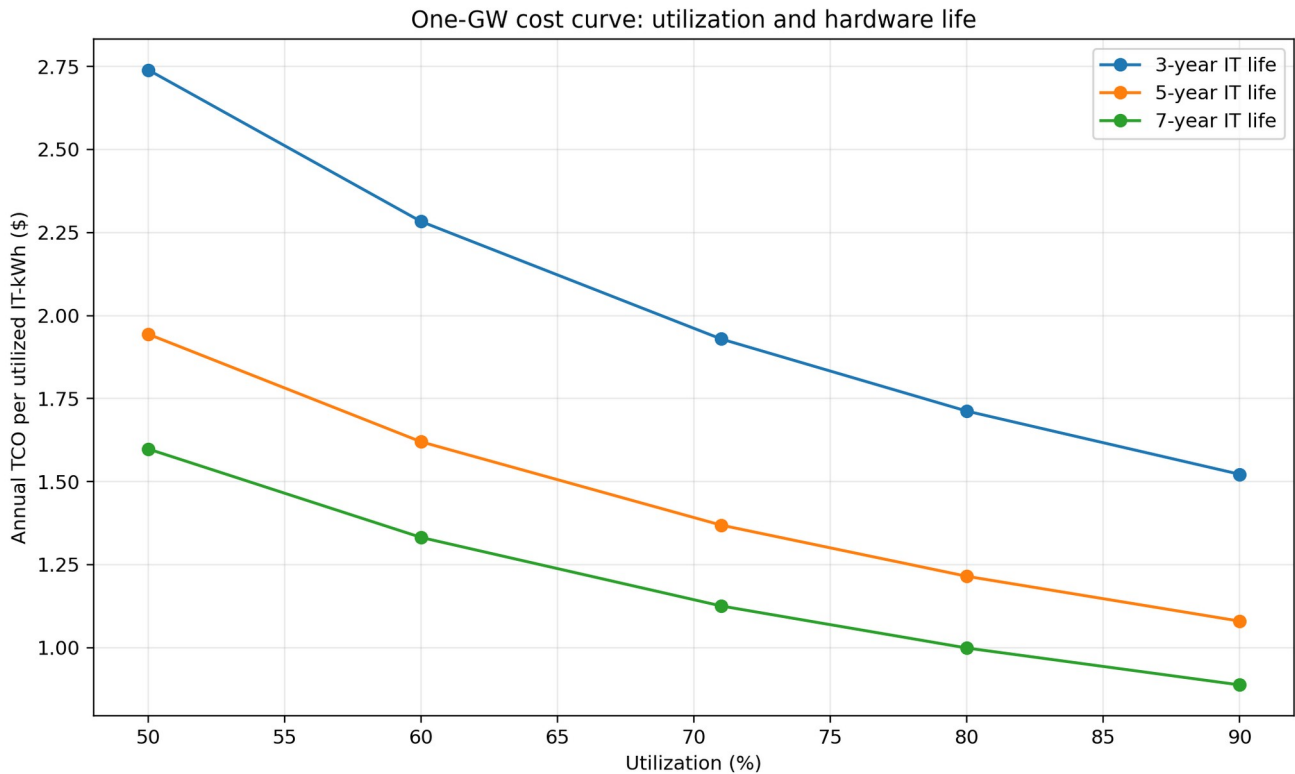


Figure 7. Cost per utilized IT-kWh versus utilization and IT asset life. [S03; Model].

Utilization	5-year cost / utilized IT-kWh	5-year cost / H100e-hour	Price for 25% operating margin
50.0%	\$1.94	\$1.41	\$1.88
60.0%	\$1.62	\$1.18	\$1.57
71.0%	\$1.37	\$0.99	\$1.32
80.0%	\$1.21	\$0.88	\$1.18

Utilization	5-year cost / utilized IT-kWh	5-year cost / H100e-hour	Price for 25% operating margin
90.0%	\$1.08	\$0.78	\$1.04

At 71% utilization, the base model costs approximately \$1.37 per utilized IT-kWh and \$0.99 per GB200-density H100e-hour. A 25% operating margin requires approximately \$1.32/H100e-hour before corporate SG&A, taxes and financing. [S03,S09-S11; Model]

8.3 Power-price sensitivity

Electricity rate	Annual energy	Annual TCO	Cost / utilized IT-kWh
\$0.0500	\$0.37B	\$8.29B	\$1.33
\$0.0834	\$0.62B	\$8.54B	\$1.37
\$0.1200	\$0.90B	\$8.82B	\$1.42
\$0.2000	\$1.49B	\$9.41B	\$1.51

Raising electricity from \$0.05 to \$0.20/kWh increases annual TCO by approximately \$1.12B/GW at 71% utilization and 1.20 PUE. Changing IT asset life from seven to three years changes annual TCO by approximately \$5B. [S03; Model]

8.4 Price-to-cost reading

Layer / example	Price	Interpretation
Powered shell	\$1.4B-\$2.3B/GW-year	Add electricity, IT hardware, operations and financing [S37,S39-S44; Model]
Full GPU service - IREN/Microsoft	\$9.7B/GW-year	Approximately 14% above \$8.514B base TCO before corporate costs [S03,S17; Model]
Oracle/OpenAI rough ratio	\$10.9B/GW-year	Approximately 28% above base TCO; cross-announcement [S03,S33,S34; Model]
Amazon/Anthropic hybrid	>\$2.0B/GW-year	Minimum value divided by maximum capacity; not comparable [S24; Model]

9. Equity-market implications

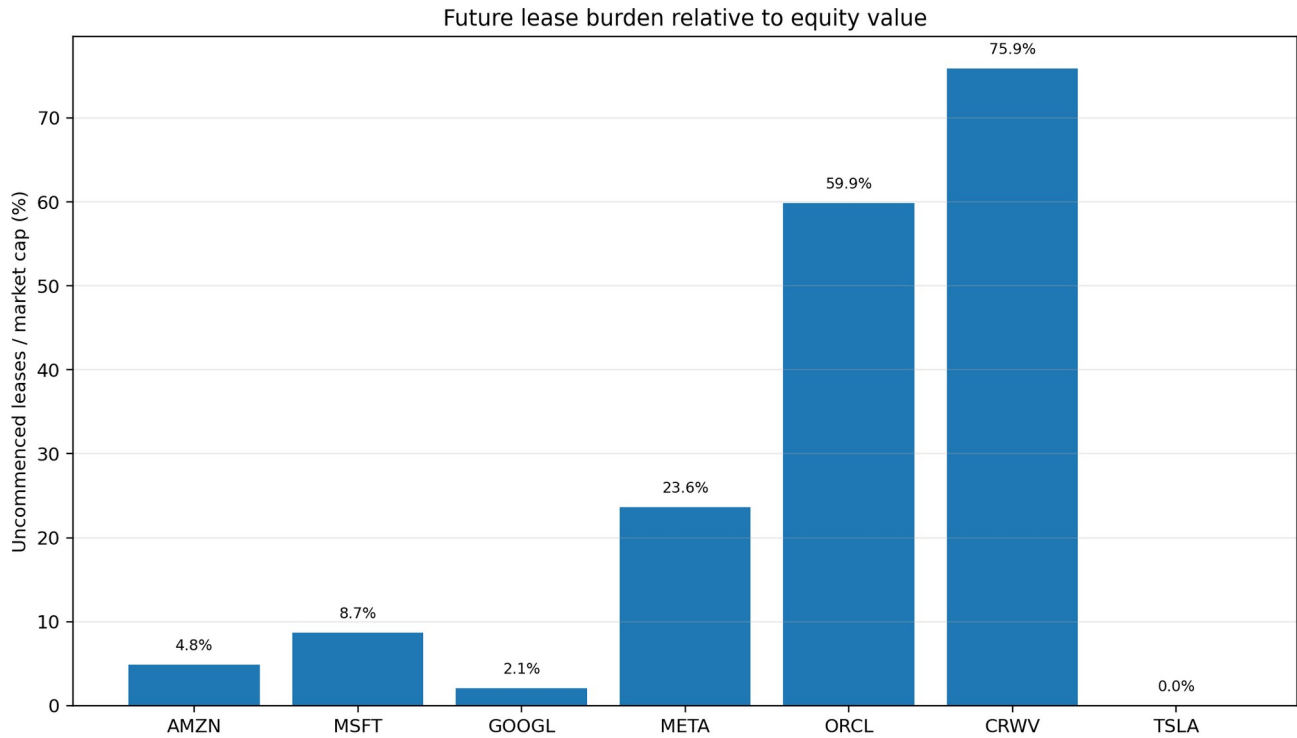


Figure 8. Uncommenced leases relative to market capitalization. [S12,S20,S23,S27,S31,S37,S48; Model].

Ticker	Close	Market cap	P/E	Current IT GW	MCap / IT GW	MCap / H100e	1 GW capex / MCap	RPO / MCap
AMZN	\$259.77	\$2,832.3B	20.9x	2.69	\$1,054.3B	\$1,151,142	1.3%	17.5%
MSFT	\$507.29	\$3,776.8B	30.2x	3.30	\$1,143.1B	\$1,077,706	1.0%	18.1%
GOOGL	\$339.35	\$4,150.3B	17.0x	4.72	\$880.2B	\$823,819	0.9%	12.5%
META	\$572.34	\$1,468.6B	21.6x	2.38	\$616.6B	\$546,567	2.6%	n/d
ORCL	\$149.12	\$434.2B	26.8x	0.49	\$880.8B	\$731,039	8.7%	146.9%
CRWV	\$84.89	\$46.8B	n/d	1.50	\$31.2B	\$28,009	81.0%	222.3%
TSLA	\$367.95	\$1,302.5B	340.7x	0.20	\$6,353.9B	\$5,663,230	2.9%	n/d

Market data [S48]; compute and RPO [S01,S04,S05,S12,S20,S23,S27,S31,S32,S37,S38,S47; Model].

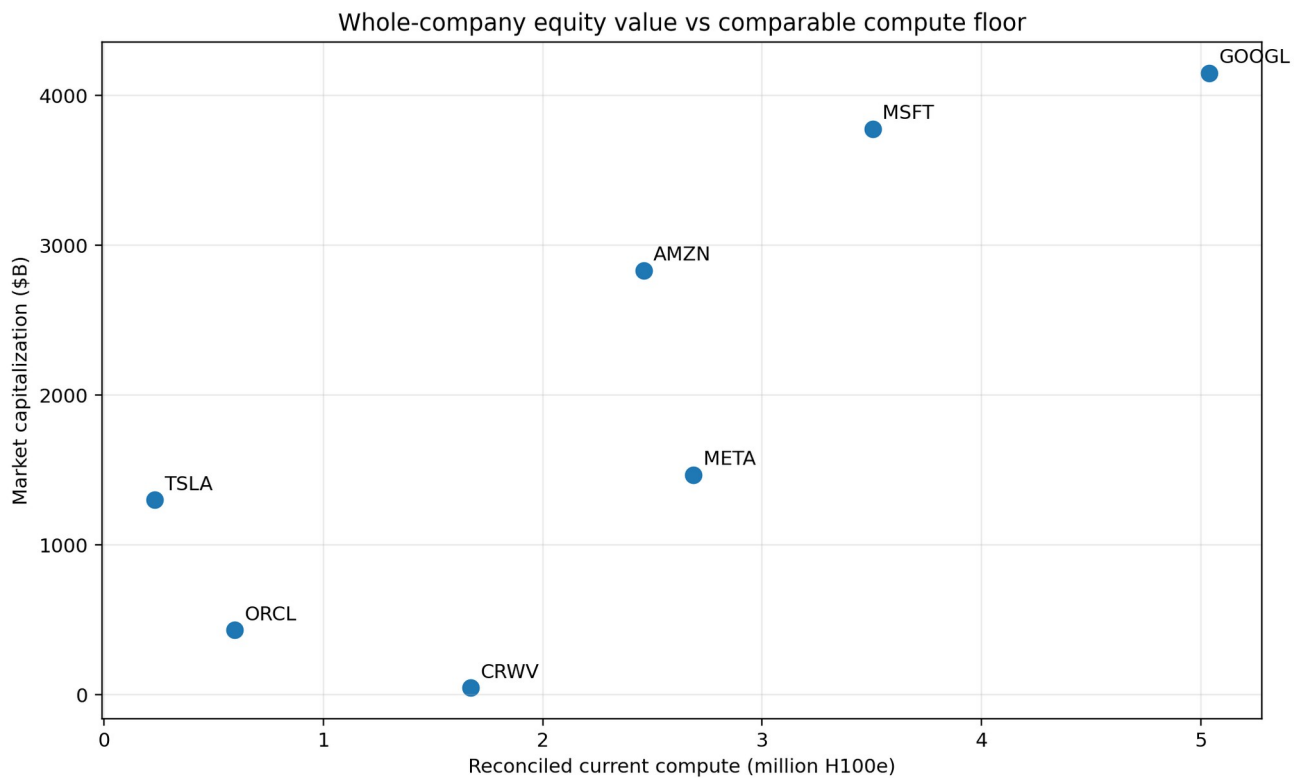


Figure 9. Whole-company equity value versus compute floor; not a data-center appraisal. [S01,S04,S05,S37,S47,S48; Model].

9.1 Hyperscalers

Alphabet has the best headline combination: the largest modeled compute fleet and the lowest P/E among Amazon, Microsoft, Alphabet and Meta. Market cap per H100e is approximately \$824,000, versus \$1.08M for Microsoft and \$1.15M for Amazon. Meta is lower at about \$547,000 because its site floor is large relative to market cap, but capacity is internally monetized. [S01,S04,S05,S48; Model]

Microsoft deserves a premium for RPO visibility and enterprise distribution, but its 30.2x P/E and \$329.1B uncommenced leases require strong utilization. Amazon has a lower 20.9x P/E and the strongest explicit vertical anchor but weaker near-term cash conversion. Meta's ads cash flows fund the build, while lease and residual commitments make its capital structure less asset-light than headline capex suggests. [S12,S23,S27,S48]

9.2 Oracle and CoreWeave

One modeled installed GW equals 8.7% of Oracle market cap and 81.0% of CoreWeave market cap, versus approximately 0.9%-2.6% for the hyperscalers. Incremental capacity therefore has much greater equity beta at Oracle and CoreWeave. [S03,S31,S37,S48; Model]

Oracle RPO is 147% of market cap and CoreWeave backlog 222%; these are exceptional demand indicators. Oracle uncommenced leases are approximately 60% of market cap, while CoreWeave debt and uncommenced leases are each approximately 76%. [S31,S32,S37,S38,S48; Model]

9.3 Scorecard

Ticker	Primary thesis	Demand / compute support	Funding / execution risk	Key variable
GOOGL	Compute leader / value	High	Moderate	TPU utilization and opaque commitments
MSFT	Demand visibility / premium	High	Moderate-high	Premium multiple and outsourced margin
AMZN	Custom silicon / anchor demand	High	Moderate	Cash conversion and ramp
META	Internal AI monetization	Medium-high	Moderate-high	No external price validation; leases/JV

Ticker	Primary thesis	Demand / compute support	Funding / execution risk	Key variable
ORCL	Highest contracted torque	Very high	Very high	Funding, customer and delivery concentration
CRWV	Pure-play compute beta	Very high	Very high	Debt, leases, concentration and refresh
TSLA	Strategic optionality	Low as asset support	High valuation sensitivity	AI outcomes beyond physical capacity

Research conclusion. GOOGL has the strongest current compute-to-valuation profile; MSFT the best broad cloud visibility; AMZN the strongest vertically integrated named anchor; META a strong funding engine without external tariff validation; ORCL the highest upside and financing torque; CRWV the purest and most levered public compute claim; TSLA compute is strategic option value rather than asset support. [S01-S05,S12,S20,S23,S27,S31-S38,S47,S48; Model]

10. Risks, uncertainty and what would change the conclusions

Risk	Why it matters
Coverage risk	The site database estimates 46% global coverage; older, smaller and undisclosed sites are missing. [S01,S02]
Overlap risk	Customer demand, supplier contracts and targets can describe the same capacity. [S16-S24,S33,S38]
Technology risk	H100e omits memory, interconnect, software and workload fit. TPU/Trainium economics may not translate cleanly. [S05-S11]
Power convention risk	A disclosed GW may be TDP, IT load, facility draw or utility interconnect. [S01,S02,S05]
Utilization risk	Cost per useful compute rises rapidly at low utilization. Contract announcements do not disclose steady-state use. [S03; Model]
Asset-life risk	Three-year versus seven-year IT life changes annual one-GW TCO by about \$5B. [S03]
Accounting risk	Capex, ROU additions, leases, commitments, prepayments and JVs allocate the same build differently. [S12,S20,S23,S27,S31,S37]
Counterparty risk	OpenAI and Anthropic anchor large pipelines; CoreWeave customers are concentrated. [S19,S21,S24,S33,S37,S38]
Market risk	Share prices and multiples are point-in-time at 31 August 2026. [S48]

The most useful future disclosures would be accelerator and IT MW by generation; active versus reserved capacity; lease starts by site; customer ramp and utilization; power pass-through; refresh responsibility; and cloud gross margin after depreciation and financing. Until then, ranges and layer-specific comparisons are more defensible than single-point fleet values.

Appendix A. Normalization detail

Platform	GPUs/sys	System kW	Ext. OH	PUE	H100e/GPU	IT kW/GPU	Physical/IT GW	H100e/IT GW	Physical/facility GW	H100e/facility GW	Source	Notes
H100 (DGX H100)	8	10.2	15.0%	1.20	1.00	1.466	682,012	682,012	568,343	568,343	S05,S06	H100e baseline; 15% external network/storage overhead is modeled.
H200	8	10.2	15.0%	1.20	1.00	1.466	682,012	682,012	568,343	568,343	S05,S06	Compute-only H100e treats H200 near H100; memory advantage is not captured.
B100 (modeled air-cooled envelope)	8	10.2	15.0%	1.20	1.77	1.466	682,012	1,206,186	568,343	1,005,155	S05,S06,S09	Uses H100 DGX power envelope as explicit approximation; dense-equivalent FP8 ratio.
B200 (DGX B200)	8	14.3	15.0%	1.20	2.27	2.056	486,470	1,106,172	405,392	921,810	S05,S07,S08	Eight-GPU DGX B200; 15% external network/storage overhead modeled.
GB200 NVL72	72	120.0	10.0%	1.18	2.53	1.833	545,455	1,378,106	462,250	1,167,887	S05,S09,S10,S11	120 kW rack plus 10% external overhead; dense FP8 equivalent.

Formula: IT kW/GPU = system kW / GPUs per system x (1 + external overhead). Facility values divide by PUE. B100 power uses an explicit H100-compatible air-cooled approximation. [S05-S11; Model]

Appendix B. Current site-level table

Vendor roll-up	Owner / customer	Site	IT MW	H100e	Modeled cost	H100e k / IT MW	Cost basis	Source
Alphabet / Google	Google	New Albany	453	616k	\$17.16B	1.36	Epoch modeled full-stack replacement capex	S01,S02
Alphabet / Google	Google	Pryor North	368	763k	\$13.94B	2.07	Epoch modeled full-stack replacement capex	S01,S02
Alphabet / Google	Google	Columbus	303	409k	\$11.48B	1.35	Epoch modeled full-stack replacement capex	S01,S02
Alphabet / Google	Google	Bristow	238	239k	\$9.02B	1.01	Epoch modeled full-stack replacement capex	S01,S02
Alphabet / Google	Google	Council Bluffs East	237	289k	\$8.98B	1.22	Epoch modeled full-stack replacement capex	S01,S02
Alphabet / Google	Google	Omaha	237	268k	\$8.98B	1.13	Epoch modeled full-stack replacement capex	S01,S02
Alphabet / Google	Google	Papillion	237	80k	\$8.98B	0.34	Epoch modeled full-stack replacement capex	S01,S02
Alphabet / Google	Google	Mesa	183	343k	\$6.93B	1.87	Epoch modeled full-stack replacement capex	S01,S02
Alphabet / Google	Google	Storey	161	154k	\$6.10B	0.95	Epoch modeled full-stack replacement capex	S01,S02
Alphabet / Google	Google	The Dalles	154	202k	\$5.83B	1.31	Epoch modeled full-stack replacement capex	S01,S02
Alphabet / Google	Google	Lincoln	141	288k	\$5.34B	2.04	Epoch modeled full-stack replacement capex	S01,S02
Alphabet / Google	Google	Lancaster	137	130k	\$5.19B	0.95	Epoch modeled full-stack replacement capex	S01,S02
Alphabet / Google	Google	Kansas City East	105	245k	\$3.98B	2.33	Epoch modeled full-stack replacement capex	S01,S02
Alphabet / Google	Google	Midlothian	103	191k	\$3.90B	1.85	Epoch modeled full-stack replacement capex	S01,S02
Alphabet / Google	Google	Waltham Cross	88	124k	\$3.33B	1.41	Epoch modeled full-stack replacement capex	S01,S02
Alphabet / Google	Google	Arcola	77	157k	\$2.92B	2.04	Epoch modeled full-stack replacement capex	S01,S02
Alphabet / Google	Google	Red Oak	77	108k	\$2.92B	1.40	Epoch modeled full-stack replacement capex	S01,S02
Amazon / AWS	Amazon / Anthropic	New Carlisle	910	686k	\$34.47B	0.75	Epoch modeled full-stack replacement capex	S01,S02
Amazon / AWS	Amazon	Madison	284	214k	\$10.76B	0.75	Epoch modeled full-stack replacement capex	S01,S02
Amazon / AWS	Amazon	Ridgeland	228	171k	\$8.64B	0.75	Epoch modeled full-stack replacement capex	S01,S02
Amazon / AWS	Amazon	New Albany	213	201k	\$8.07B	0.94	Epoch modeled full-stack replacement capex	S01,S02
Amazon / AWS	Amazon	Berwick	108	81k	\$4.09B	0.75	Epoch modeled full-stack replacement capex	S01,S02
Anthropic	Anthropic / Fluidstack	Lake Mariner	42	59k	\$1.59B	1.40	Epoch modeled full-stack replacement capex	S01,S02

Vendor roll-up	Owner / customer	Site	IT MW	H100e	Modeled cost	H100e k / IT MW	Cost basis	Source
CoreWeave	CoreWeave	Denton	262	253k	\$9.93B	0.96	Epoch modeled full-stack replacement capex	S01,S02
CoreWeave	CoreWeave	Helios	132	160k	\$5.00B	1.21	Epoch modeled full-stack replacement capex	S01,S02
CoreWeave	CoreWeave	Chester	82	88k	\$3.11B	1.08	Epoch modeled full-stack replacement capex	S01,S02
CoreWeave	CoreWeave	Ellendale	68	94k	\$2.58B	1.38	Epoch modeled full-stack replacement capex	S01,S02
CoreWeave	CoreWeave	Marble	65	79k	\$2.46B	1.21	Epoch modeled full-stack replacement capex	S01,S02
CoreWeave	CoreWeave	Norway	42	51k	\$1.59B	1.21	Epoch modeled full-stack replacement capex	S01,S02
CoreWeave	CoreWeave	Dalton 1 & 2	28	32k	\$1.06B	1.15	Epoch modeled full-stack replacement capex	S01,S02
Meta	Meta	Prometheus	562	680k	\$21.29B	1.21	Epoch modeled full-stack replacement capex	S01,S02
Meta	Meta / QTS	Hillsboro 2	180	124k	\$6.82B	0.69	Epoch modeled full-stack replacement capex	S01,S02
Meta	Meta	Jeffersonville	178	203k	\$6.74B	1.14	Epoch modeled full-stack replacement capex	S01,S02
Meta	Meta	Rosemount	178	216k	\$6.74B	1.21	Epoch modeled full-stack replacement capex	S01,S02
Meta	Meta	Montgomery	153	174k	\$5.80B	1.14	Epoch modeled full-stack replacement capex	S01,S02
Meta	Meta	Cheyenne	152	173k	\$5.76B	1.14	Epoch modeled full-stack replacement capex	S01,S02
Meta	Meta	Kuna	152	173k	\$5.76B	1.14	Epoch modeled full-stack replacement capex	S01,S02
Meta	Meta	Temple	152	173k	\$5.76B	1.14	Epoch modeled full-stack replacement capex	S01,S02
Meta	Meta	Huntsville	146	167k	\$5.53B	1.14	Epoch modeled full-stack replacement capex	S01,S02
Meta	Meta	Aiken	142	162k	\$5.38B	1.14	Epoch modeled full-stack replacement capex	S01,S02
Meta	Meta	Eagle Mountain	133	151k	\$5.04B	1.14	Epoch modeled full-stack replacement capex	S01,S02
Meta	Meta	Gallatin	87	100k	\$3.30B	1.14	Epoch modeled full-stack replacement capex	S01,S02
Meta	Meta	Los Lunas	87	100k	\$3.30B	1.14	Epoch modeled full-stack replacement capex	S01,S02
Meta	Meta	Sarpy	80	92k	\$3.03B	1.15	Epoch modeled full-stack replacement capex	S01,S02
Microsoft	Microsoft	Fairwater Atlanta	636	769k	\$24.09B	1.21	Epoch modeled full-stack replacement capex	S01,S02
Microsoft	Microsoft	Fairwater Wisconsin	369	446k	\$13.98B	1.21	Epoch modeled full-stack replacement capex	S01,S02
Microsoft	Microsoft	Goodyear	202	205k	\$7.65B	1.02	Epoch modeled full-stack replacement capex	S01,S02

Vendor roll-up	Owner / customer	Site	IT MW	H100e	Modeled cost	H100e k / IT MW	Cost basis	Source
Microsoft	Microsoft	Project Osmium	190	156k	\$7.20B	0.82	Epoch modeled full-stack replacement capex	S01,S02
Microsoft	Microsoft	SAT40	75	65k	\$2.84B	0.86	Epoch modeled full-stack replacement capex	S01,S02
Microsoft	Microsoft / Nebius	New Jersey	50	57k	\$1.89B	1.14	Epoch modeled full-stack replacement capex	S01,S02
Microsoft	Microsoft	SAT14	47	45k	\$1.78B	0.96	Epoch modeled full-stack replacement capex	S01,S02
Oracle	Oracle / OpenAI	Stargate Abilene	421	509k	\$15.95B	1.21	Epoch modeled full-stack replacement capex	S01,S02
Oracle	Oracle	Batam	72	84k	\$2.73B	1.17	Epoch modeled full-stack replacement capex	S01,S02
xAI / SpaceXAI	xAI / SpaceXAI	Colossus 2	946	1,112k	\$35.84B	1.18	Epoch modeled full-stack replacement capex	S01,S02
xAI / SpaceXAI	xAI	QTS Atlanta	17	12k	\$0.66B	0.66	Epoch modeled full-stack replacement capex	S01,S02

The appendix contains the 55 major-vendor current rows used in the report, not all 86 campuses in the underlying database. Modeled cost is a common replacement-cost estimate. [S01,S02]

Appendix C. Planned / target site-level table

Vendor roll-up	Owner / sponsor	Site	Target IT MW	Matched current MW	Incremental MW	Target H100e	Incremental H100e	Modeled cost	Source
Alphabet / Google	Google	Fort Wayne	1,035	0	1,035	3,600k	3,600k	\$32.69B	S01,S02
Alphabet / Google	Google	Omaha	474	237	237	1,100k	832k	\$14.96B	S01,S02
Alphabet / Google	Google	Columbus	464	303	161	707k	299k	\$14.62B	S01,S02
Alphabet / Google	Google	Storey	389	161	228	652k	498k	\$11.36B	S01,S02
Alphabet / Google	Google	Lincoln	339	141	198	869k	582k	\$10.72B	S01,S02
Alphabet / Google	Google	Cedar Rapids	314	0	314	816k	816k	\$9.89B	S01,S02
Alphabet / Google	Google	Red Oak	185	77	108	316k	208k	\$5.83B	S01,S02
Amazon / AWS	Amazon / Anthropic	New Carlisle	2,310	910	1,400	1,700k	1,014k	\$72.92B	S01,S02
Amazon / AWS	Amazon	Ridgeland	1,008	228	780	788k	617k	\$31.82B	S01,S02
Amazon / AWS	Amazon	Madison	819	284	535	514k	300k	\$25.87B	S01,S02
Amazon / AWS	Amazon	New Albany	405	213	192	352k	151k	\$12.80B	S01,S02
Amazon / AWS	Amazon	Berwick	212	108	104	145k	64k	\$6.40B	S01,S02
Anthropic	Anthropic / Fluidstack	Lake Mariner	441	42	399	527k	468k	\$14.32B	S01,S02
Anthropic	Anthropic	Barber Lake	300	0	300	288k	288k	\$7.84B	S01,S02
CoreWeave	CoreWeave	Helios	800	132	668	638k	479k	\$20.00B	S01,S02
CoreWeave	CoreWeave	Ellendale	520	68	452	505k	412k	\$15.15B	S01,S02
CoreWeave	CoreWeave	Denton	395	262	133	272k	19k	\$10.68B	S01,S02
CoreWeave	CoreWeave	Chester	197	82	115	126k	37k	\$4.28B	S01,S02
CoreWeave	CoreWeave	Muskogee	181	0	181	198k	198k	\$5.80B	S01,S02
Crusoe	Crusoe	Abilene expansion	941	0	941	1,700k	1,700k	\$25.46B	S01,S02
Meta	Meta	Hyperion	2,262	0	2,262	4,300k	4,300k	\$63.49B	S01,S02
Meta	Meta	Prometheus	1,110	562	548	1,000k	320k	\$32.35B	S01,S02
Meta	Meta	Montgomery	465	153	312	408k	233k	\$13.56B	S01,S02
Meta	Meta	Cheyenne	264	152	112	251k	78k	\$8.33B	S01,S02
Meta	Meta	Temple	231	152	79	202k	29k	\$6.74B	S01,S02
Microsoft	Microsoft	Fairwater Wisconsin	2,715	369	2,346	4,500k	4,054k	\$85.73B	S01,S02
Microsoft	Microsoft	Narvik	124	0	124	265k	265k	\$3.49B	S01,S02
OpenAI	OpenAI	Stargate New Mexico	2,450	0	2,450	5,100k	5,100k	\$66.29B	S01,S02
OpenAI	OpenAI	Stargate Shackelford	1,960	0	1,960	4,600k	4,600k	\$53.03B	S01,S02
OpenAI	OpenAI	Stargate UAE	1,400	0	1,400	2,000k	2,000k	\$37.88B	S01,S02
OpenAI	OpenAI	Stargate Michigan	1,383	0	1,383	2,900k	2,900k	\$37.43B	S01,S02
OpenAI	OpenAI	Stargate Wisconsin	1,300	0	1,300	2,600k	2,600k	\$34.17B	S01,S02

Vendor roll-up	Owner / sponsor	Site	Target IT MW	Matched current MW	Incremental MW	Target H100e	Incremental H100e	Modeled cost	Source
OpenAI	OpenAI	Stargate Milam	1,200	0	1,200	2,500k	2,500k	\$32.46B	S01,S02
OpenAI	OpenAI	Lordstown	300	0	300	546k	546k	\$8.11B	S01,S02
Oracle	Oracle / OpenAI	Stargate Abilene	1,180	421	759	1,000k	491k	\$31.93B	S01,S02
QTS	QTS	Cedar Rapids	1,587	0	1,587	3,700k	3,700k	\$42.92B	S01,S02
xAI / SpaceXAI	xAI / SpaceXAI	Colossus 2	2,065	946	1,119	1,800k	688k	\$58.00B	S01,S02

Incremental capacity subtracts an exact same-name current site within the same vendor roll-up. It does not prove signing, financing, interconnect completion or final delivery. [S01,S02; Model]

Appendix D. Source register

ID	Category	Issuer	Document / dataset	Date	URL	Use
S01	Independent dataset	Epoch AI	AI Data Centers dataset	2026-08-31	https://epoch.ai/data/ai-data-centers	Named current/planned site IT power, H100e, modeled capex, coverage and uncertainty.
S02	Methodology	Epoch AI	AI Data Centers methodology	2026-08-31	https://epoch.ai/data/data-centers-documentation/methodology	Definitions, estimation methods, common cost convention and uncertainty.
S03	Cost model	Epoch AI	AI data center cost breakdown	2025-06-12	https://epoch.ai/data-insights/ai-datacenter-cost-breakdown	One-GW installed cost and annual TCO component model.
S04	Independent dataset	Epoch AI	AI Chip Owners dataset	2026-08-31	https://epoch.ai/data/ai-chip-owners	Modeled accelerator stocks by owner and chip type.
S05	Methodology	Epoch AI	AI Chip Owners methodology	2026-08-31	https://epoch.ai/data/ai-chip-owners-documentation/methodology	H100e methodology and accelerator-TDP ownership model.
S06	Technical specification	NVIDIA	DGX H100 user guide - introduction	Current	https://docs.nvidia.com/dgx/dgxh100-user-guide/introduction-to-dgxh100.html	Eight H100 GPUs and 10.2 kW maximum system power.
S07	Technical specification	NVIDIA	DGX B200	Current	https://www.nvidia.com/en-us/data-center/dgx-b200/	Eight B200 GPUs, 14.3 kW maximum power and FP8 throughput.
S08	Technical specification	NVIDIA	HGX platform	Current	https://www.nvidia.com/en-gb/data-center/hgx/	B200 eight-GPU platform and FP8 throughput.
S09	Technical specification	NVIDIA	Blackwell Architecture Technical Overview	Current	https://resources.nvidia.com/en-us-blackwell-architecture	B100 throughput/power and GB200 platform performance.
S10	Technical specification	NVIDIA	DGX GB200 hardware	Current	https://docs.nvidia.com/dgx/dgxgb200-user-guide/hardware.html	NVL72 configuration: 72 Blackwell GPUs and 36 Grace CPUs.
S11	Technical specification	NVIDIA	Mission Control FAQ - GB200 rack power	Current	https://docs.nvidia.com/mission-control/docs/systems-administration-guide/2.0.0/prs/faq.html	Approximately 120 kW full-load GB200 NVL72 rack power.

ID	Category	Issuer	Document / dataset	Date	URL	Use
S12	SEC filing	Microsoft	FY2026 Form 10-K	2026-07-29	https://microsoft.gcs-web.com/static-files/8a9beeed-0e9e-48e6-b4fb-593d558e2693	PPE, leases, purchase/construction commitments, RPO and cash flow.
S13	Company announcement	Microsoft	Fairwater Wisconsin architecture	2025-09-18	https://blogs.microsoft.com/blog/2025/09/18/inside-the-worlds-most-powerful-ai-datacenter/	Campus acreage, area and hundreds of thousands of GB200 GPUs.
S14	Company announcement	Microsoft	First Fairwater Wisconsin facility completed	2026-06-23	https://news.microsoft.com/source/2026/06/23/microsoft-completes-construction-on-first-datacenter-facility-in-mount-pleasant-wisconsin/	Operational timing and \$4.7B local construction spending estimate.
S15	Company announcement	Microsoft	Azure AI superfactory - Fairwater Atlanta	2025-11-12	https://blogs.microsoft.com/blog/2025/11/12/infinite-scale-the-architecture-behind-the-azure-ai-superfactory/	Hundreds of thousands of GB200/GB300 and approximately 140 kW rack design.
S16	Counterparty announcement	Nscale	Nscale and Microsoft capacity agreements	2025	https://www.nscale.com/press-releases/nscale-microsoft-2025	Approximately 200,000 GB300; Texas 104,000/~240 MW; other sites and 700 MW option.
S17	SEC filing	IREN	8-K: Microsoft AI cloud contract	2025-11-03	https://www.sec.gov/Archives/edgar/data/1878848/000114036125040072/ef20058139_8k.htm	\$9.7B/5-year, 200 MW IT, 20% prepayment and \$5.8B Dell GPU capex.
S18	SEC filing	Nebius	6-K: Microsoft GPU services agreement	2025-09-08	https://www.sec.gov/Archives/edgar/data/1513845/000110465925088312/tm2525580d1_6k.htm	\$17.4B through 2031, option to \$19.4B; capacity undisclosed.

ID	Category	Issuer	Document / dataset	Date	URL	Use
S19	Company announcement	Microsoft	Microsoft, NVIDIA and Anthropic partnerships	2025-11-18	https://blogs.microsoft.com/blog/2025/11/18/microsoft-nvidia-and-anthropic-announce-strategic-partnerships/	Anthropic \$30B Azure commitment and up to 1 GW additional capacity.
S20	SEC filing	Alphabet	Q2 2026 Form 10-Q	2026-07-24	https://www.sec.gov/Archives/edgar/data/1652044/000165204426000071/goog-20260630.htm	Capex, PPE, assets not in service, leases, commitments, guarantees, debt and RPO.
S21	Customer announcement	Anthropic	Expanding use of Google Cloud TPUs	2025-10-23	https://www.anthropic.com/news/expanding-our-use-of-google-cloud-tpus-and-services	Up to 1M TPUs, tens of billions and well over 1 GW online in 2026.
S22	Customer announcement	Anthropic	Higher usage limits and compute deals	2026-05-06	https://www.anthropic.com/news/higher-limits-spacex	Confirms 5 GW Google/Broadcom compute program beginning in 2027; overlap undisclosed.
S23	SEC filing	Amazon	Q2 2026 Form 10-Q	2026-07-31	https://www.sec.gov/Archives/edgar/data/1018724/000101872426000026/amzn-20260630.htm	Cash capex, AWS PPE additions/RPO, leases, purchase obligations and commitments.
S24	Customer announcement	Anthropic	Anthropic and Amazon expand compute partnership	2026-04-20	https://www.anthropic.com/news/anthropic-amazon-compute	>\$100B/10 years, up to 5 GW new, nearly 1 GW by end-2026 and >1M Trainium2.
S25	Company announcement	Amazon	Project Rainier	2025-06-24	https://www.aboutamazon.com/news/aws/aws-project-rainier-ai-trainium-chips-compute-cluster	Nearly 500k Trainium2 initially; expected >1M.
S26	Company announcement	Amazon	\$15B Indiana data-center investment	2026	https://www.aboutamazon.com/news/company-news/amazon-15-billion-indiana-data-centers	\$15B campus investment and 2.4 GW power envelope.

ID	Category	Issuer	Document / dataset	Date	URL	Use
S27	SEC filing	Meta	Q2 2026 Form 10-Q	2026-07-30	https://www.sec.gov/Archives/edgar/data/1326801/000162828026050705/meta-20260630.htm	Capex, OCF, PPE/CIP, leases, commitments, debt, restricted cash and capacity contingencies.
S28	Company announcement	Meta	Hyperion expansion in Louisiana	2026-07	https://about.fb.com/news/2026/07/teachers-local-businesses-win-as-meta-expands-louisiana-data-center/	5 GW compute target and >\$50B regional investment.
S29	Company announcement	Meta	Blue Owl joint venture for Hyperion	2025	https://investor.atmeta.com/investor-news/press-release-details/2025/Meta-Announces-Joint-Venture-with-Funds-Managed-by-Blue-Owl-Capital-to-Develop-Hyperion-Data-Center/default.aspx	80/20 JV, approximately \$27B assets, Meta lease and residual-value guarantee.
S30	Company announcement	Meta	Nuclear energy projects	2026-01	https://about.fb.com/news/2026/01/meta-nuclear-energy-projects-power-american-ai-leadership/	Up to 6.6 GW nuclear support by 2035; energy sourcing, not compute.
S31	SEC filing	Oracle	FY2026 Form 10-K	2026-06-26	https://www.sec.gov/Archives/edgar/data/1341439/000119312526277521/orcl-20260531.htm	Capex, OCF, prepayments, PPE/CIP, leases, obligations, guarantees and debt.
S32	Company results	Oracle	FY2026 results	2026-06-10	https://investor.oracle.com/investor-news/news-details/2026/Oracle-Announces-Record-Q4-and-FY-2026-Results-Driven-by-Cloud-Infrastructure--Cloud-Applications/default.aspx	Revenue, cloud revenue, capital funding and \$638B RPO.

ID	Category	Issuer	Document / dataset	Date	URL	Use
S33	Customer announcement	OpenAI	Stargate advances with Oracle	2025-07-22	https://openai.com/index/stargate-advances-with-partnership-with-oracle/	4.5 GW additional Oracle capacity, >5 GW under development and >2M chips.
S34	Customer announcement	OpenAI	Five new Stargate sites	2025-09-23	https://openai.com/index/five-new-stargate-sites/	Nearly 7 GW, >\$400B/3 years and Oracle partnership >\$300B / >5.5 GW.
S35	Customer announcement	OpenAI	Expanding Stargate to Michigan	2026	https://openai.com/index/expanding-stargate-to-michigan/	>1 GW Michigan; Stargate >8 GW planned and >\$450B/3 years.
S36	Customer announcement	OpenAI	Announcing the Stargate Project	2025-01-21	https://openai.com/index/announcing-the-stargate-project/	Original \$500B / 10 GW ambition.
S37	SEC filing	CoreWeave	Q2 2026 Form 10-Q	2026-08-12	https://www.sec.gov/Archives/edgar/data/1769628/000176962826000366/crww-20260630.htm	PPE, ROU assets, cash/noncash capex, leases, debt, interest and customer concentration.
S38	Company results	CoreWeave	Q2 2026 results	2026-08-12	https://investors.coreweave.com/news/news-details/2026/CoreWeave-Reports-Strong-Second-Quarter-2026-Results/default.aspx	1.5 GW active, 3.7 GW contracted, approximately \$104B backlog and earnings metrics.
S39	Counterparty announcement	Core Scientific	Core Scientific/CoreWeave 590 MW expansion	2025	https://investors.corescientific.com/news-events/press-releases/detail/110/core-scientific-and-coreweave-announce-1-2-billion-expansion-at-denton-tx-site	590 MW critical IT and \$10.2B aggregate 12-year lease value.

ID	Category	Issuer	Document / dataset	Date	URL	Use
S40	Counterparty filing	Core Scientific	Q1 2025 earnings materials	2025	https://investors.corescientific.com/sec-filings/all-sec-filings/content/0001628280-25-023161/exhibit992-q12025earning.htm	Approximately \$850M annual lease revenue, tenant-paid power/buildout and approximately \$1.5M/MW credit.
S41	Counterparty filing	Core Scientific	Q4/FY2025 earnings deck	2026	https://investors.corescientific.com/sec-filings/all-sec-filings/content/0001628280-26-013214/q4fy25earningsdeckvfinal.htm	Critical-IT MW schedule by site.
S42	Counterparty announcement	Galaxy	Helios Phase I completion	2026	https://www.galaxy.com/newsroom/galaxy-completes-phase-i-of-its-helios-data-center-campus	526 MW critical/800 MW gross, 15-year lease and >\$1B annual revenue.
S43	Counterparty announcement	Applied Digital	Polaris Forge 1 Phase 1 ready for service	2026	https://ir.applieddigital.com/news-events/press-releases/detail/133/applied-digital-achieves-ready-for-service-for-phase-1-at	400 MW CoreWeave lease, \$11B and approximately 15 years.
S44	Counterparty announcement	Applied Digital	\$5B AI factory lease	2026	https://ir.applieddigital.com/news-events/press-releases/detail/132/applied-digital-announces-5-billion-ai-factory-lease-with	200 MW, \$5B and 15 years.
S45	Counterparty results	Applied Digital	FY2026 results	2026	https://ir.applieddigital.com/news-events/press-releases/detail/159/applied-digital-reports-fiscal-fourth-quarter-and-full-year	Additional 300 MW/\$7.5B and 210 MW/\$5.2B leases.

ID	Category	Issuer	Document / dataset	Date	URL	Use
S46	Counterparty presentation	Applied Digital	Investor presentation	2026	https://ir.applieddigital.com/sec-filings/all-sec-filings/content/0001144879-26-000036/apld_invxinvestorpresent.htm	New-build capex \$11M-\$13M per critical-IT MW; power passed through.
S47	Company update	Tesla	Q2 2026 shareholder update	2026-07-22	https://ir.tesla.com/flysystem/s3/sec/000162828026049213/tsla-20260722-gen.pdf	Cortex 1 >90 MW, Cortex 2 >115 MW, visual roadmap, capex and cash flow.
S48	Market data	Market-data feed	US equity snapshot	2026-08-31	finance://AMZN,MSFT,GO,ORCL,META,ORCL,CRWV,TSLA/2026-08-31	Latest close, market capitalization and trailing P/E.

All sources were retrieved or verified for this report on 1 September 2026. S48 records the finance snapshot by ticker and date.